

Heterogeneous Edge Clients

Client 1

Tier 1 · low-end

...

Client i

Tier 2 · mid-range

...

Client N

Tier 3 · high-end

local updates
+ state $s_{i,t}$

broadcast θ_t
to Top-K

Central Server — MAML-Select

Meta-Policy

h_ϕ

MLP 6-64-64-1

Inner Look-Ahead

$$\phi'_t = \phi_t - \beta \nabla \mathcal{L}_{\text{sup}}$$

Score & Select Top-K

$$\arg \text{TopK}[-h_{\phi'_t}(s_{i,t})]$$

**Aggregate &
Outer Update**

$$\phi_{t+1} = \phi_t - \eta \nabla \mathcal{L}_{\text{qry}}$$

policy carried to round $t+1$

state $s_{i,t}$: training loss · gradient norm · latency · battery · selection frequency · staleness